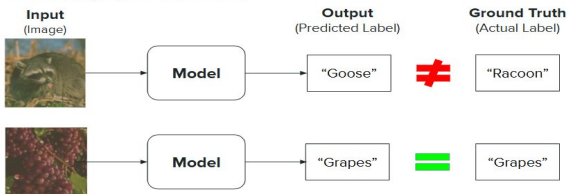


Multi-Class Image Classification with Caltech-256

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Problem Statement

- Enhance **single-label image classification** by utilizing pixel data through a **Hybrid-CNN** model, applying the **Model Soup** strategy, and incorporating **ensemble classifiers**, while integrating **Vision Transformers (ViT)** to boost performance.



Baseline ViT - Motivation

Token-Based Attention

- Transformers process sequences of tokens via self-attention.

Self-Attention Computation

- For each token, the model computes three vectors: Query (Q), Key (K), and Value (V). The dk is the dimension of K vector.
- Then it calculates the Attention(Q,K,V): $\text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$
- FLOPs: $O(\text{length}^2 \cdot \text{dim})$
- Length: number of patches & Dim: embedding dimension

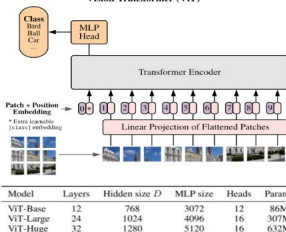
Computational Bottleneck:

- Self-attention requires pairwise interactions.
- NLP Tasks require approx 500 tokens which will require $500^2 = 250K$ attention connections.
- Image Tasks may need $224 \times 224 = 50,176$ pixels which will require $50,176^2 = \text{approx } 2.52B$ attention connections.
- Directly treating each pixel as a token is Impractical.

An Image is Worth 16x16 Words

Paper - Google Research [1]

Vision Transformer (ViT)



Single-Image Classification Dataset - Caltech 256 [2]

Number of Classes: 256 + 1 (clutter class) = 257

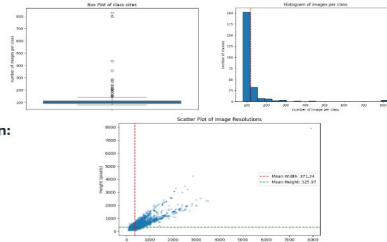
Total Images: Approx 30,607

Limitations:

- Class Imbalance:
- Highly Variable Resolution:

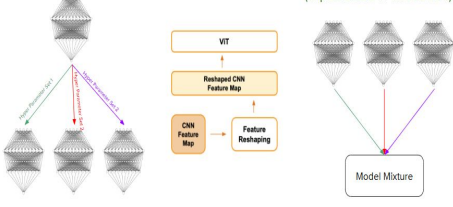
Solutions:

- Weighted Sampling
- Weighted Loss



Proposed Updates

- Basic Fine-Tuning (Implemented in Milestone I)
- ViT-CNN Hybrid Model [3] (Implemented in Milestone I)
- Greedy, Diversity, and Uniform Model Soup [4] (Implemented in Milestone II)



Proposed Updates: Building a Dual-Attention GridMix for Enhanced Augmentation for Fine-Tuning

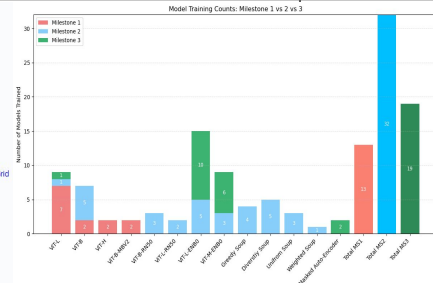
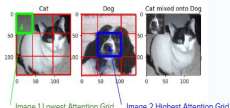
(Implemented in Milestone II)

Dual-Attention GridMix (Our own implementation):

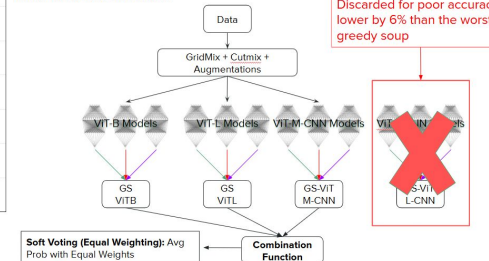
- Dividing images into a mxn grid.
- Extracts attention maps from ViT's last layer attention heads.
- Aggregate scores per patch forming a mxn attention grid.
- Identified Highest & Lowest attention grid cell in 2 Images.
- For image 1 select Highest attention grid cell from image 2 and paste it on top of the lowest attention grid cell on image 1

Advantage:

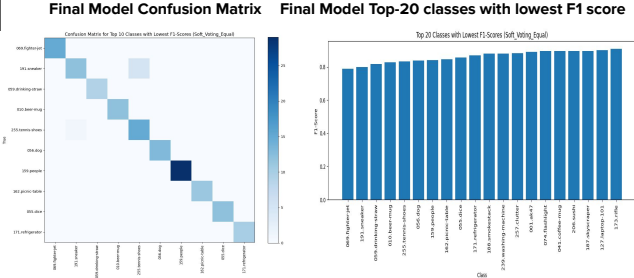
- Avoids the standard Attention Cutmix flow of overlapping high attention regions



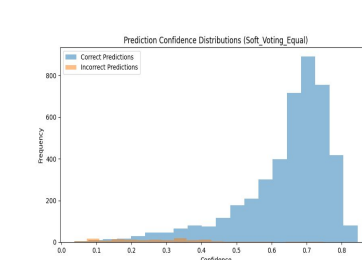
Final Model Architecture



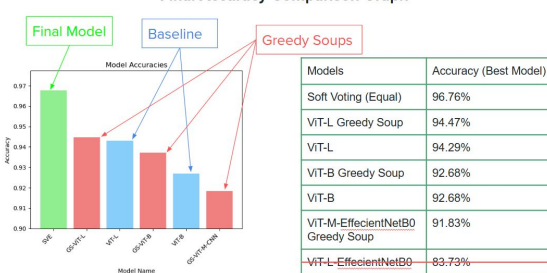
Final Model Confusion Matrix



Final Model Prediction Confidence Distribution



Final Accuracy Comparison Graph



ViT-Explainability (XAI)

- Examples of Heatmaps on some classes on ViT-B Model N3.

